ADAPTIVE COGNITIVE-LOAD STUDY TIMER FOR PERSONALIZED BREAK SCHEDULING

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Project Proposal Report

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Department of Information Technology
Sri Lanka Institute of Information Technology
Sri Lanka

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Author Details

- Author Name – Rajendram P. A.
- Student ID No. – IT22087874
- Research objective – Intent-Lock Overlay for Distraction Prevention

Supervisor Details

- Supervisor Name – Dr. kalpani Manathunga.
- Co-Supervisor Name – mr eishan

Declaration

I declare that this is my own work and this proposal does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgment is made in the text.

Name	Student ID	Signature
Rajendram P. A.	IT22087874	

Statement of the supervisor,

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the supervisor: Date: 22/08/2024

Abstract

Maintaining sustained focus during study sessions is often disrupted by impulsive task-switching, where students exit learning applications for unrelated activities. Conventional distraction prevention tools either rigidly block access, frustrating users with genuine needs, or allow unrestricted exits, enabling frequent loss of focus. To address these limitations, this project introduces the Intent-Lock Overlay, an adaptive intervention mechanism that balances flexibility and accountability while promoting reflective self-control.

The proposed system intercepts exit attempts and applies graduated interventions based on user behavior: soft nudges such as reminders and progress statistics, stronger friction through confirmation prompts, and adaptive interventions that escalate according to patterns of distraction. A lightweight predictive model (e.g., logistic regression or decision tree) is employed to distinguish between impulsive and genuine exits, using features such as time since session start, typing pace variations, and frequency of app switches. In addition, reflective prompts capture distraction reasons, which are categorized to build a personalized taxonomy of focus vulnerabilities.

The methodology involves designing the overlay interface, implementing intervention strategies, integrating predictive classification, and evaluating the system with pilot user studies to assess usability, adaptability, and impact on focus stability. Expected results include reduced impulsive exits, improved awareness of distraction patterns, and the generation of unique behavioral metrics—such as distraction attempt frequency and confirmation time—that serve both user feedback and research purposes.

By moving beyond rigid blocking toward adaptive, data-informed reflection, this study component contributes a novel approach to distraction prevention, empowering students to sustain productive study sessions while respecting genuine breaks.

Table of Contents

Supplementary Sections

List of Figures

List of Tables

List of Abbreviations

Abbreviation	**Full Form**
API	Application Programming Interface
CHI	Conference on Human Factors in Computing Systems
DSCT	Digital Self-Control Tools
EMA	Ecological Momentary Assessment
HCI	Human–Computer Interaction
PNAS	Proceedings of the National Academy of Sciences
SDK	Software Development Kit
LR	Logistic Regression
DT	Decision Tree

1. Introduction

Students often face difficulties maintaining sustained focus during study sessions, with impulsive distractions such as checking social media or switching to unrelated apps disrupting their concentration. Traditional distraction prevention tools, such as rigid app blockers or simple timers, provide limited effectiveness—either frustrating users with strict lockouts or offering no resistance to impulsive task-switching. These approaches lack adaptability and fail to account for the difference between a genuine need to exit and an impulsive urge.

To address this challenge, this research proposes the Intent-Lock Overlay, an adaptive focus-preserving mechanism that introduces graduated friction when users attempt to leave study mode. Instead of outright blocking or freely allowing distractions, the overlay intervenes intelligently: offering gentle nudges, requiring confirmation steps, or escalating to stronger prompts based on user behavior and distraction frequency. A predictive model is also incorporated to classify exit attempts as impulsive or genuine, while reflective prompts capture distraction reasons to build personalized distraction profiles.

The goal of this project is to create a tool that not only prevents impulsive exits but also enhances user awareness by tracking distraction patterns, providing feedback, and supporting sustainable focus during study sessions.

1.1 Background and Literature Survey

Digital distractions represent one of the most pervasive challenges to maintaining focus in academic contexts. Students frequently engage in impulsive task-switching—such as opening social media or unrelated applications during study sessions—which leads to reduced concentration, fragmented attention, and lower learning efficiency. Traditional focus and productivity applications typically adopt a binary approach: either blocking access rigidly or allowing distractions freely. While rigid blocking may reduce impulsivity, it often frustrates users with genuine needs and reduces adoption; conversely, permissive systems provide little real protection against distraction. This highlights the need for adaptive systems that balance flexibility with accountability.

Research in **digital self-control tools (DSCTs)** has emphasized the role of reflection and intentional friction as effective strategies for reducing impulsive technology use. Mark et al. (2018) demonstrated that introducing subtle friction such as requiring users to pause before accessing distracting applications—significantly reduced impulsive digital behavior, compared to rigid blocking approaches [1]. Similarly, Lyngs et al. (2019) provided a taxonomy of DSCTs, highlighting that reflection-based interventions can enhance user autonomy while still effectively mitigating distractions [2]. These findings suggest that intentional friction mechanisms are more sustainable than punitive approaches in distraction prevention.

Recent experimental studies further validate reflection-based interventions. The **"one sec" app study** published in *PNAS* (Matthies et al., 2023) found that requiring users to reflect on their intent before opening a distracting app reduced usage frequency by up to 57% [3]. This aligns with broader HCI research demonstrating the effectiveness of "friction nudges" in promoting deliberate decision-making over impulsive actions (Kim et al., 2019) [4].

Beyond reflection, advances in **predictive modeling** enable systems to distinguish between *impulsive* and *genuine* exit attempts. Logistic regression and decision tree classifiers have been widely used in behavior prediction tasks due to their interpretability and low computational overhead (D'Mello & Graesser, 2015) [5]. In cognitive load and attention research, lightweight models analyzing features such as typing pace, session timing, and app-switch frequency have been shown to effectively capture shifts in engagement (Zhou et al., 2020) [6]. Applying these models to distraction prevention enables systems to intelligently adapt interventions based on the type of exit attempt.

Another underexplored area is the **logging and taxonomy of distraction reasons**. Research on self-regulation emphasizes that awareness of one's triggers is essential for sustainable behavior change (Duckworth et al., 2016) [7]. By capturing reasons such as fatigue, boredom, or curiosity, and categorizing them into a personalized distraction taxonomy, systems can help users identify their vulnerability patterns—an aspect missing in most current tools.

Despite these advances, existing applications and studies rarely integrate reflection, predictive classification, and distraction logging into a single adaptive framework. Current DSCTs often rely on either static friction or basic time-blocking, neglecting personalization and data-driven adaptation. This project aims to bridge this gap by developing the **Intent-Lock Overlay**, a system that (a) introduces adaptive, graduated interventions, (b) applies predictive models to classify impulsive versus genuine exits, and (c) builds distraction taxonomies to enhance user self-awareness. This integration represents a novel contribution to the domain of digital distraction prevention and learning technologies.

1.2 Research Gap

While distraction management has been studied extensively in psychology and HCI, most existing digital self-control tools (DSCTs) and productivity applications fail to integrate these insights into practical, adaptive systems. Current applications such as **Forest, Freedom, and StayFocusd** adopt rigid blocking mechanisms, which either frustrate users by denying legitimate needs or provide no flexibility to differentiate between impulsive and genuine exits [1][2]. Similarly, timer-based applications such as

Pomodoro apps only enforce static time intervals and do not address the underlying problem of impulsive task-switching.

Although reflection-based interventions such as the *“one sec”* app have demonstrated measurable reductions in impulsive app use [3], existing solutions lack predictive intelligence to determine whether an exit attempt is impulsive or intentional. Furthermore, most DSCTs do not record or analyze the reasons for distraction, leaving users without meaningful insights into their personal vulnerability patterns. This omission limits long-term self-awareness and prevents the design of tailored interventions.

Another significant gap lies in the absence of adaptive friction mechanisms. While research highlights the effectiveness of friction nudges and reflective prompts [4][5], commercial systems generally apply uniform interventions regardless of context or user behavior. There is little to no incorporation of machine learning methods such as logistic regression or decision trees, which could enable lightweight classification of impulsive versus genuine exits based on behavioral features like session length, typing pace variations, or switch frequency [6].

In summary, existing distraction prevention tools suffer from three key limitations:

- **Rigid or static interventions** that fail to balance user autonomy with effective focus-preservation.
- **Lack of predictive intelligence** to distinguish impulsive exits from genuine needs.
- **No taxonomy of distraction reasons**, preventing deeper personalization and reflection.

This research addresses these gaps by introducing the **Intent-Lock Overlay**, which integrates adaptive friction strategies, predictive exit classification, and distraction taxonomy building. By combining these elements, the system not only prevents impulsive task-switching but also generates actionable insights into individual distraction patterns—a feature missing in current tools.

1.3 Research Problem

The primary problem this research addresses is the **lack of adaptive, intelligent distraction prevention mechanisms** that balance user autonomy with effective focus-preservation. Current tools either enforce rigid lockouts or apply uniform interventions, overlooking the difference between an impulsive distraction and a genuine exit need. As a result, students frequently experience either **excessive frustration** from being overly restricted or **unmitigated loss of focus** from insufficient intervention.

Existing applications also fail to integrate **predictive modeling** to classify exit attempts. Without this capability, systems treat all exit attempts equally, missing the opportunity to apply nuanced interventions. Moreover, the absence of **distraction reason logging and taxonomy building** prevents students from gaining awareness of their recurring triggers, such as fatigue, boredom, or curiosity, which are essential for long-term self-regulation [7].

Therefore, the research problem can be summarized as follows:

- **How can we design an adaptive distraction prevention system that introduces personalized friction, distinguishes between impulsive and genuine exit attempts, and builds user-specific distraction profiles, thereby reducing impulsive task-switching while supporting sustainable focus?**

This study proposes the **Intent-Lock Overlay** as a solution: a system that (1) adaptively introduces graduated interventions based on user behavior, (2) employs lightweight predictive models to classify exit attempts, and (3) records distraction reasons to construct personalized taxonomies. By integrating these approaches, the system seeks to overcome the limitations of existing DSCTs and provide a sustainable, user-centered method for distraction prevention in educational contexts.

2. Objectives

2.1 Main Objectives

The main objective of this module is to design and implement an intent-lock overlay that introduces reflective friction when users attempt to disengage from study sessions. Instead of either rigidly blocking or freely allowing distractions, the overlay applies adaptive interventions that encourage users to pause, reflect, and consciously decide before exiting. By integrating graduated friction levels, predictive detection of impulsive versus genuine exits, and reflection prompts, the module aims to minimize impulsive task-switching while respecting genuine study breaks. Furthermore, it seeks to build a personalized distraction profile by logging disengagement reasons and behavioral patterns, thereby contributing to long-term self-regulation, enhanced focus stability, and improved learning outcomes.

2.2 Specific Objectives

1. **Design and implement an intent-lock overlay with graduated intervention levels.**

   - **Friction Modes:** Develop soft, firm, and adaptive intervention modes that introduce intentional pauses, reminders, or effort-based actions before allowing distraction exits.

   - **Dynamic Control:** Enable escalation of intervention strength when repeated distraction attempts are detected within the same session.

2. **Integrate predictive modeling for impulsive versus genuine exit classification:**

    - **Model Selection:** Evaluate lightweight models such as Logistic Regression and Decision Trees to classify exit attempts.

    - **Feature Engineering:** Incorporate contextual features such as session duration, app-switch frequency, typing pace drops, and cognitive load signals from the system.

3. **Implement reflection-based disengagement loggin:**

    - **Reason Capture:** Prompt users to record disengagement reasons (e.g., fatigue, boredom, social media urge, urgent task).

    - **Profile Building:** Aggregate logged reasons into a personalized distraction profile for long-term behavioral insights.

4. **Evaluate the effectiveness of the overlay on reducing impulsive task-switching:**

    - **Pilot Testing:** Conduct trials with students to measure impulsive-exit reduction, intervention acceptance, and overall usability.

    - **Adaptation:** Refine intervention thresholds and predictive model parameters based on real-world pilot feedback.

5. **Measure and document system performance:**

- **Performance Metrics:** Track distraction attempts, bypass latency, intervention effectiveness, and user satisfaction.

- **Documentation:** Prepare module-specific documentation covering design, algorithm choices, user guidelines, and integration pathways for system deployment and future improvements.

3. Methodology

This module focuses on preventing impulsive disengagements during study sessions by introducing reflective friction when users attempt to access distractions. The system first intercepts exit attempts and analyzes contextual features such as session duration, typing pace, and app-switch frequency. A lightweight predictive model then classifies the attempt as impulsive or genuine, enabling the overlay to apply adaptive interventions ranging from soft reminders to firm confirmations. Reflection prompts are used to capture user reasons for disengagement, which are stored in a distraction profile for long-term analysis. These insights help the system dynamically adapt interventions while contributing to a deeper understanding of users' focus patterns.

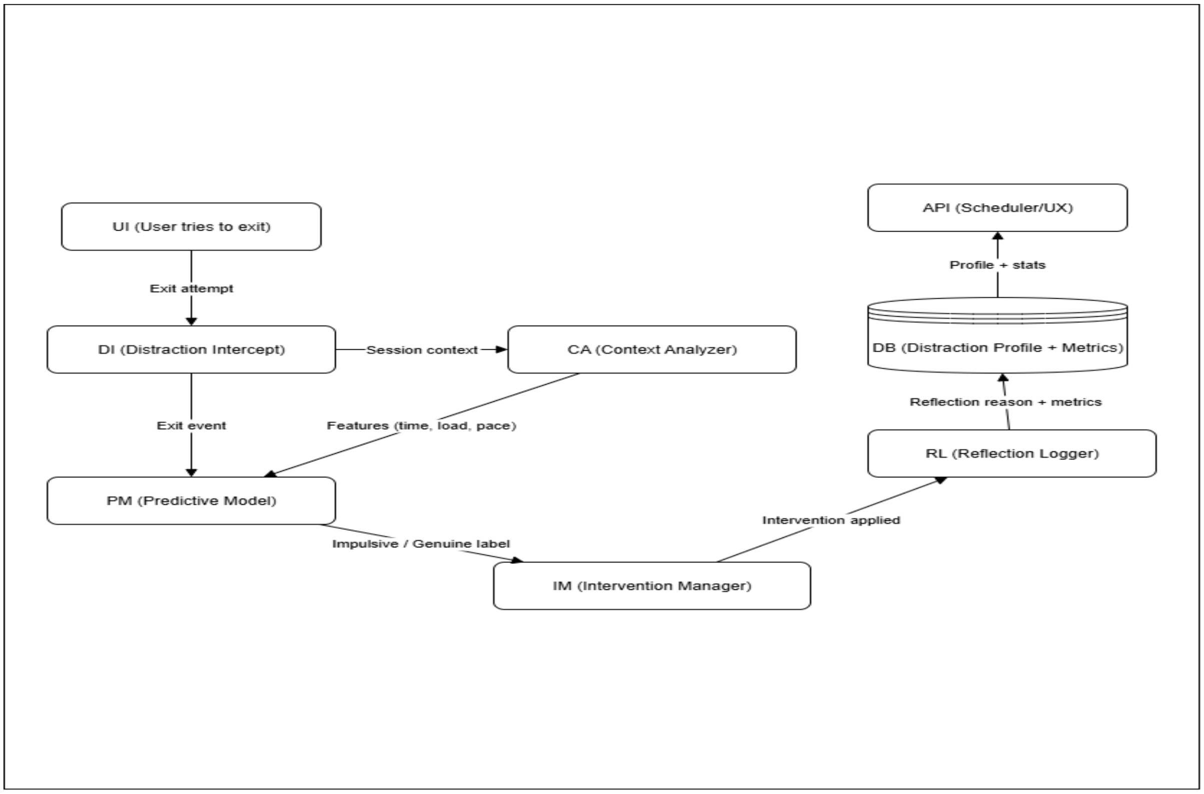

Figure 3. 1 – system diagram of Intent Lock Overlay Component

3.1 Overall System Description

1. **System Architecture:**

   - **Distraction Intercept Layer:** Captures user attempts to leave the study session or switch to distracting applications. It serves as the entry point for intent-lock interventions.

   - **Adaptive Intervention Manager:** Applies graduated friction mechanisms (soft nudges, firm confirmations, adaptive escalation) based on the context and frequency of distraction attempts.

- **Predictive Exit Classifier:** A lightweight model (Logistic Regression or Decision Tree) processes contextual features (time in session, typing pace, app-switch frequency) to classify exit attempts as impulsive or genuine.

- **Reflection and Logging Unit:** Prompts users to provide disengagement reasons and records bypass latency, distraction attempts, and intervention outcomes. Logged data feeds into a personalized distraction profile for long-term analysis.

2. **Software Components:**

- **Context Analyzer:** Preprocesses real-time data such as session duration, typing pace drops, and app-switch behavior to provide contextual features for the predictive model.

- **Predictive Model Engine:** Implements impulsive vs. genuine exit classification using trained lightweight ML models and passes results to the intervention manager.

- **Reflection Logger**: Captures user-reported disengagement reasons (e.g., fatigue, boredom, social media) and stores them in the distraction profile database.

- **Backend Integration:** Handles data storage for distraction profiles, bypass latency, and metrics while ensuring communication with the main database and other cognitive-load modules.

3.2 Implementation Steps

1. **Exit Attempt Capture & Context Preprocessing:**
    - **Intercept User Actions:** Monitor user attempts to exit study mode or switch to distracting apps.
    - **Feature Extraction:** Collect contextual features such as session duration, typing pace variations, app-switch frequency, and time of day.
2. **Predictive Exit Classification:**
    - **Model Selection:** Implement lightweight classifiers such as Logistic Regression or Decision Trees for exit classification.
    - **Model Training:** Train the model using pilot study data where exits are labeled as impulsive or genuine based on reflection feedback.
3. **Adaptive Intervention Manager:**
    - **Friction Mechanisms:** Implement graduated interventions (soft nudges, firm confirmations, adaptive escalation).
    - **Dynamic Adjustment:** Increase intervention strictness if repeated distraction attempts occur within a single session.

4. **Reflection and Logging Module:**

    - **Reason Collection:** Prompt users to provide disengagement reasons (e.g., boredom, fatigue, curiosity, urgent task).
    - **Profile Generation:** Store logged reasons, bypass latency, and intervention outcomes in a distraction profile database.

5. **Integration with Study Timer System:**

    - **APIs and Data Flow:** Ensure smooth communication with the cognitive load timer, passing distraction metrics and receiving load signals.

    - **Unified Dashboard:** Enable visualization of distraction attempts and focus stability alongside study session progress.

6. **Testing and Evaluation:**

    - **System Testing:** Conduct unit, integration, and functional testing of the overlay module.

    - **User Evaluation:** Test with real students to measure impulsive-exit reduction, intervention effectiveness, and impact on overall study efficiency.

4. Project Requirements

4.1 Functional Requirements

Hardware Components:

- **Mobile/PC Devices:**
    - Smartphones (Android/iOS) or PCs: Devices used by users to interact with the study timer and provide activity data.

Software Components:

- **Vs Code / Android Studio:**
    - **Programming Environment:** Used for writing software implementation

- **Intent-Lock Overlay:**
    - **Overlay Layer:** Detects and intercepts user attempts to leave study mode, applying friction and reflection prompts before exit is allowed.
    - **Graduated Interventions:** Implements soft nudges (reminders), firm actions (long press, typed phrase), and adaptive escalation based on repeated attempts.

- **Predictive Model Module:**

- **Exit Classifier:** Lightweight ML model (Logistic Regression or Decision Tree) to classify disengagement attempts as impulsive or genuine.

- **Reflection Logger & Database:**
    - **Distraction Profile Storage:** Logs disengagement reasons, bypass latency, and attempt frequency to build a personalized distraction profile.
    - **Integration with Study Timer:** Shares distraction metrics and insights with the broader cognitive-load system for synchronized analysis.

4.2 Non-Functional Requirements

- **Privacy & Security:**
    - All logged distraction reasons and exit data should remain on-device or securely stored, with user consent required for any cloud sync.

- **Adaptivity:**
    - The overlay must dynamically adjust intervention levels (soft → firm → adaptive) based on real-time user behavior and predictive model outputs.

- **Usability:**
    - Interventions should be minimally disruptive, allowing smooth study flow while preventing impulsive exits.

    - Reflection prompts must be quick and easy to complete, avoiding frustration.

- **Reliability:**
    - Consistently intercept exit attempts across supported devices and apps.
    - Ensure accurate classification of impulsive vs. genuine exits with minimal false detections.

- **Performance & Efficiency:**
    - The overlay should run as a lightweight background process with minimal battery and resource consumption.
    - Predictive model inference must be fast enough for real-time decision-making without delays.

4.3 Expected Test Cases

Exit Attempt Capture Testing:

- **Objective:** Verify that all user exit attempts (e.g., app-switching, closing study mode) are consistently intercepted by the overlay.
- **Procedure:** Simulate multiple exit attempts on different devices and confirm that the overlay activates each time.

Predictive Model Classification Testing:

- **Objective:** Ensure the predictive model accurately distinguishes between impulsive and genuine exits.

- **Procedure:** Feed the model with simulated session features (short vs. long sessions, frequent app-switches, typing pace changes) and check classification results against expected outcomes.

Intervention Mechanism Testing:

- **Objective:** Validate that graduated interventions (soft, firm, adaptive) are triggered correctly based on context and repeated attempts.
- **Procedure:** Attempt multiple distractions within one session and confirm escalation from reminders → long press/typed phrase → stricter interventions.

Reflection Logging Testing:

- **Objective:** Confirm accurate capture of disengagement reasons, bypass latency, and attempt frequency.
- **Procedure:** Perform exits with different reflection reasons (fatigue, boredom, urgent task) and verify correct storage in the distraction profile database.

Integration Testing:

- **Objective:** Ensure smooth communication between the overlay, predictive model, reflection logger, and the main study timer system.
- **Procedure:** Run full study sessions with distractions and check that metrics are logged, interventions occur, and distraction data is shared with the broader system.

5. Budget and Budget Justification

The budget for this project includes the costs for hardware components, software licenses, and personnel involved in the development and testing of the system. Below are the estimated costs to develop a prototype for the implementation.

Table 5. 1 - Estimated Budget Allocation

	Description	Cost
Hardware Components	Mobile Deveices and sensors	Rs 0
Software Components	Vs code	Open-source
	Android Studio	Open-source
	Libraries and SDKs	Open-source

6. Self-Evaluation Plan

To ensure quality and rigour, the following self-evaluation criteria will be used:

• **Relevance of interventions:** Do the implemented friction levels (soft, firm, adaptive) align with findings in digital well-being and HCI research? Is there evidence that they effectively reduce impulsive exits?

• **Predictive model accuracy:** Does the exit classifier achieve acceptable performance (e.g., >70% accuracy in distinguishing impulsive vs. genuine exits)? Is it robust across different users and session contexts?

• **Usability of the overlay:** Are interventions minimally disruptive while still effective? Do reflection prompts capture meaningful reasons without frustrating users? Pilot feedback will guide refinements.

• **Privacy and ethics:** Is distraction-related data stored securely on-device? Are users fully informed about what is logged (reasons, latency) and how it is used?
• **Integration readiness:** Does the overlay expose a clean API to the broader study timer system? Are latency and response times acceptable (<200 ms) when intercepting exits?
Progress will be documented weekly, with iterative adjustments based on user testing and supervisor feedback.

7. Commercialization Potential

The intent-lock overlay has strong potential for commercialization as a stand-alone feature or SDK that can be integrated into productivity and learning applications. Unlike rigid blockers, it introduces adaptive, humane interventions that balance flexibility and accountability — a differentiator in the crowded digital well-being market. Growing awareness of screen time management and self-regulation among students and professionals creates demand for such tools.

Potential commercialization pathways include:
• Integration with study apps: Embed the overlay into existing timers or e-learning platforms.
• SDK/Plugin model: Package as a modular component for third-party productivity apps.
• Institutional adoption: Offer to universities or schools as part of digital well-being initiatives.

Challenges for commercialization include demonstrating long-term user acceptance, ensuring low battery/resource consumption, and proving scalability across devices. Partnerships with educational institutions or app developers could accelerate adoption.

8. Conclusion

This proposal presents the design and development of an intent-lock overlay to reduce impulsive disengagement during study sessions. By intercepting exit attempts, applying graduated friction, and prompting reflection, the system encourages conscious decisions rather than impulsive task-switching. A lightweight predictive model classifies exits as impulsive or genuine, ensuring adaptive interventions that respect legitimate needs while discouraging distractions.

Reflection data and bypass latency are logged to build personalized distraction profiles, offering valuable insights into focus stability and habit patterns. The methodology emphasizes adaptive friction, predictive modeling, and behavioral logging to create a balanced, user-friendly defense against distractions. Successful completion will not only enhance the adaptive study timer with a novel distraction-prevention mechanism but also contribute to the broader field of digital self-regulation and human–computer interaction.

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